Transistor (BJT) Review

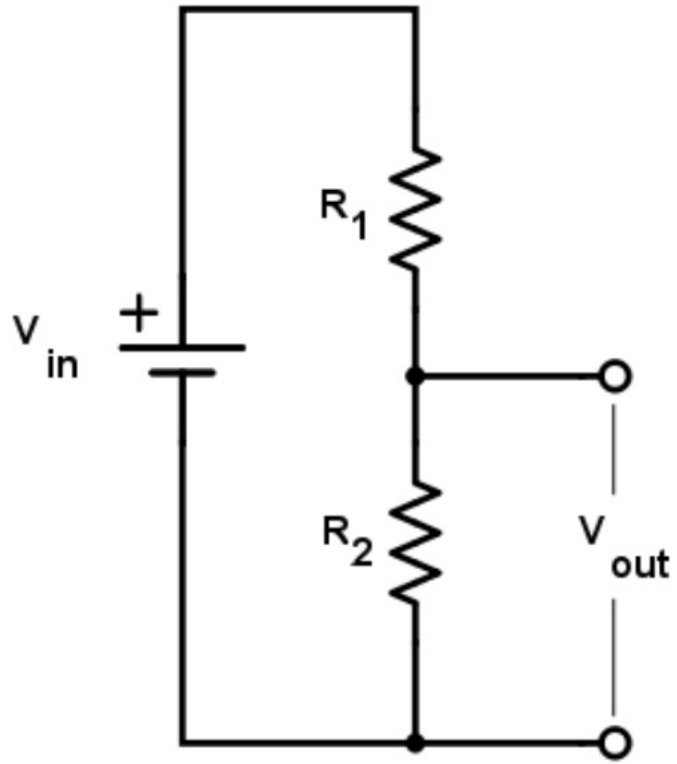
Ginanjar S. Adi

Tujuan Pembelajaran

1. Mengulas kembali materi mengenai dasar elektronika dan BJT
2. Memahami parameter dasar dalam rangkaian BJT
3. Mampu melakukan perhitungan dan mendesain rangkaian BJT sederhana penguat sinyal kecil

Quiz

1. Berapakah nilai Vout jika Vin = 10V; R1 = 3KOhm; R2 = 2KOhm. Tuliskan persamaannya! (10 Poin)

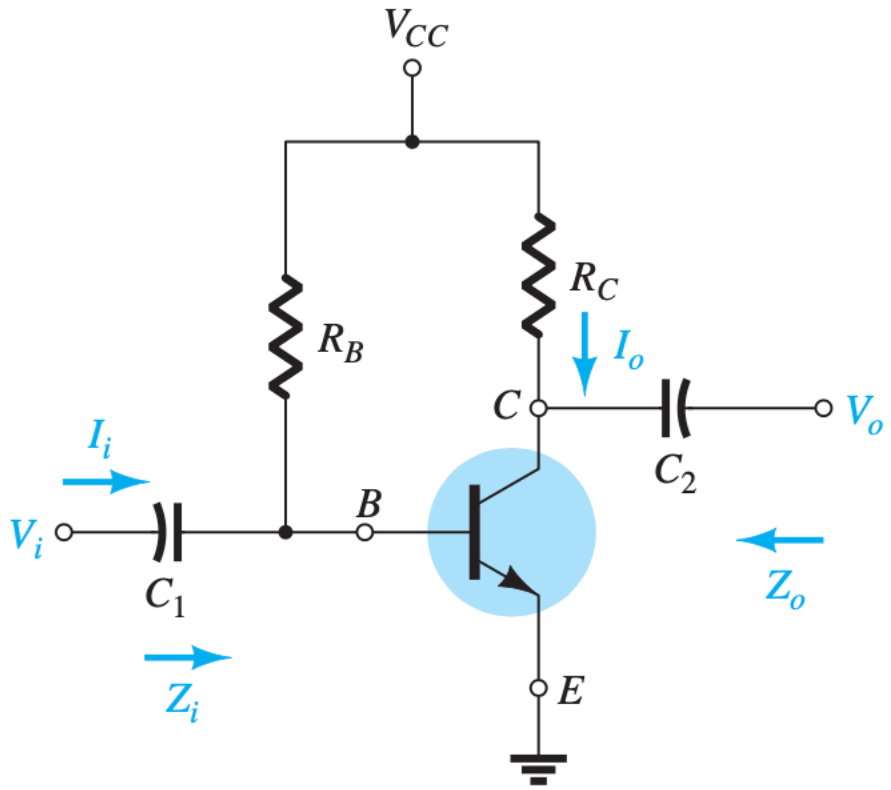


Quiz

2. Gambarkan simbol BJT tipe NPN! (5 poin)
3. Apa parameter yang paling mempengaruhi perubahan nilai I_C dalam rangkaian BJT (konfigurasi rangkaian common emitter)? Tuliskan persamaannya! (5 poin)
4. Berapakah nilai ideal tegangan V_{BE} dari suatu konfigurasi rangkaian common emitter? (5 poin)

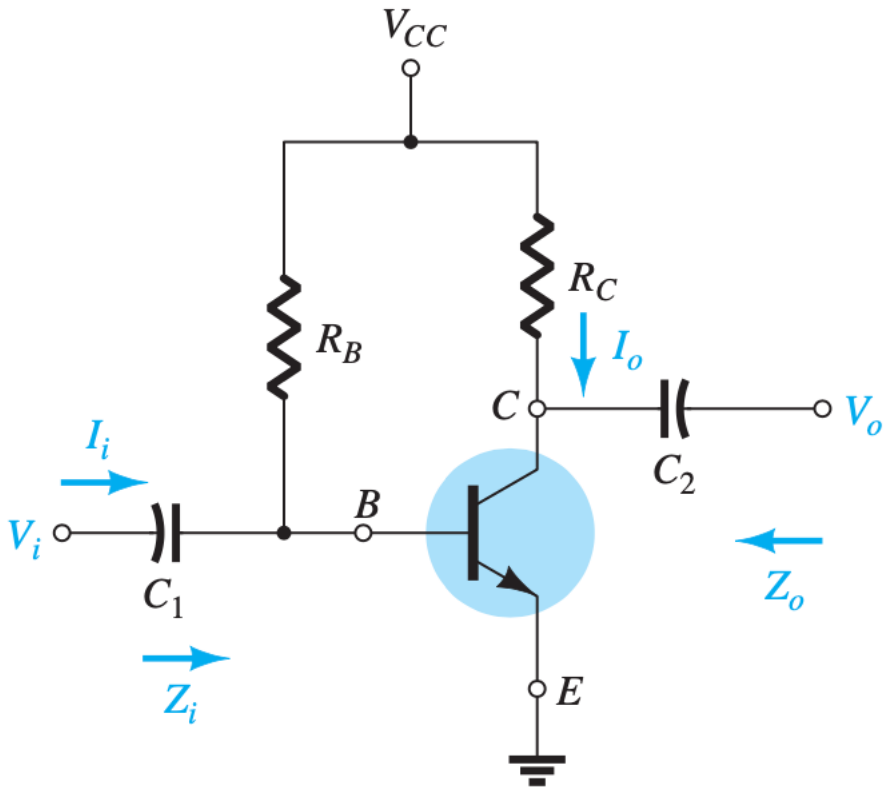
Quiz

5. Berapakah nilai X_{C1} dan X_{C2} pada analisis DC untuk rangkaian di bawah ini , jika nilai $C_1 = C_2 = 10uF$ (10 poin)



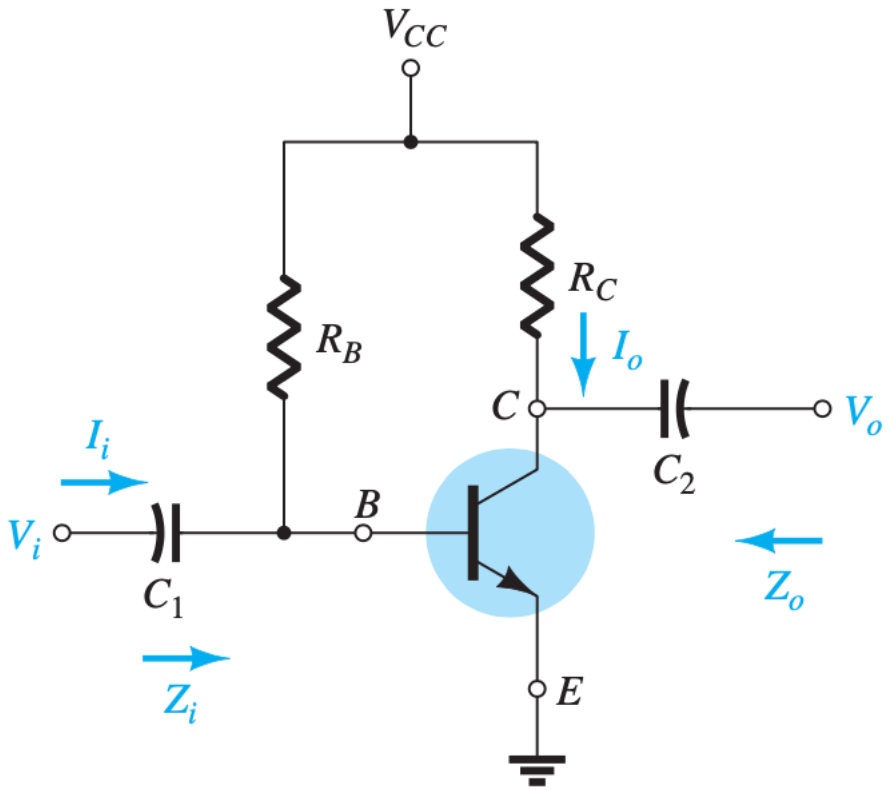
Quiz

6. Gambarkan rangkaian pengganti untuk analisis DC rangkaian di samping ini!(10 poin)
7. Gambarkan garis beban di sisi output! (5 poin)
8. Jika nilai hfe atau β = 200, Tentukan nilai V_{CC}, R_C, R_B, dan I_B untuk kondisi daerah aktif jika diinginkan nilai $I_{CQ} \cong 5mA$ dan $V_{CEQ} \cong 5V$ (dimana titik kerja tepat ditengah-tengah dari garis beban) (20 poin)



Quiz

9. Gambarkan rangkaian pengganti ekivalen untuk analisis sinyal AC dari rangkaian berikut! (10 poin)
10. Tuliskan persamaan A_V jika nilai r_o diperhitungkan dan untuk nilai $r_o \geq 10R_C$ (20 poin)



Datasheets (1/3)

BC107/BC108 Series

Low Power Bipolar Transistors

multicomp

Absolute Maximum Ratings

Description	Symbol	BC107	BC108	Unit
Collector-Emitter Voltage	V_{CEO}	45	25	V
Collector-Base Voltage	V_{CBO}	50	30	
Emitter-Base Voltage	V_{EBO}	6.0	5.0	
Collector Current Continuous	I_C	0.2		A
Power Dissipation at T_a = 25°C Derate above 25°C	P_D	0.6 2.28		W mW/°C
Power Dissipation at T_C = 25°C Derate above 25°C		1.0 6.67		
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-65 to +200		°C
Thermal Resistance				
Junction to Case	$R_{th\ (j-c)}$	175		°C/W

Datasheets (2/3)

Electrical Characteristics (T_a = 25°C unless otherwise specified)

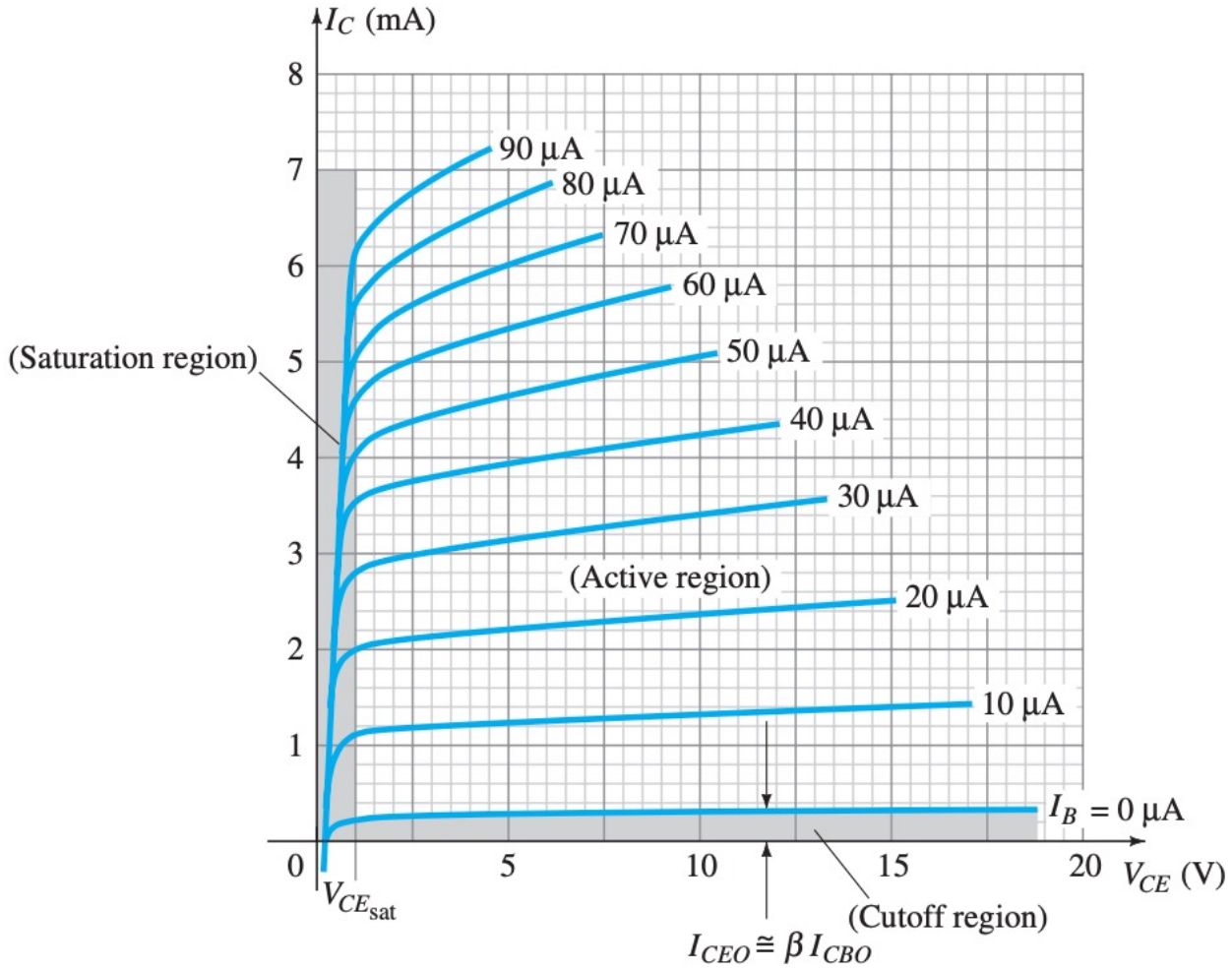
Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector-Emitter Voltage	V_{CEO}	I_C = 2mA, I_B = 0 BC107 BC108	45 25	-	V
Emitter-Base Voltage	V_{EBO}	I_E = 10µA, I_C = 0 BC107 BC108	6.0 5.0	-	V
Collector-Cut off Current	I_{CBO}	V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108 T_{amb} = 125°C V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108	-	15 15 4.0 4.0	nA µA
DC Current	h_{FE}	I_C = 10µA, V_{CE} = 5V **B Group** **C Group** I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	40 100 110 110 110 200 420	- - 450 800 220 450 800	-
Base Emitter Saturation Voltage	$V_{BE\ (Sat)}$	I_C = 10mA, I_B = 0.5mA I_C = 100mA, I_B = 5mA	-	0.83 1.05	V
Collector Emitter Saturation Voltage	$V_{CE\ (Sat)}$		-	0.25 0.60	V
Base Emitter On Voltage	$V_{BE\ (on)}$	I_C = 2mA, V_{CE} = 5V I_C = 10mA, V_{CE} = 5V	0.55 -	0.70 0.77	V

Datasheets (3/3)

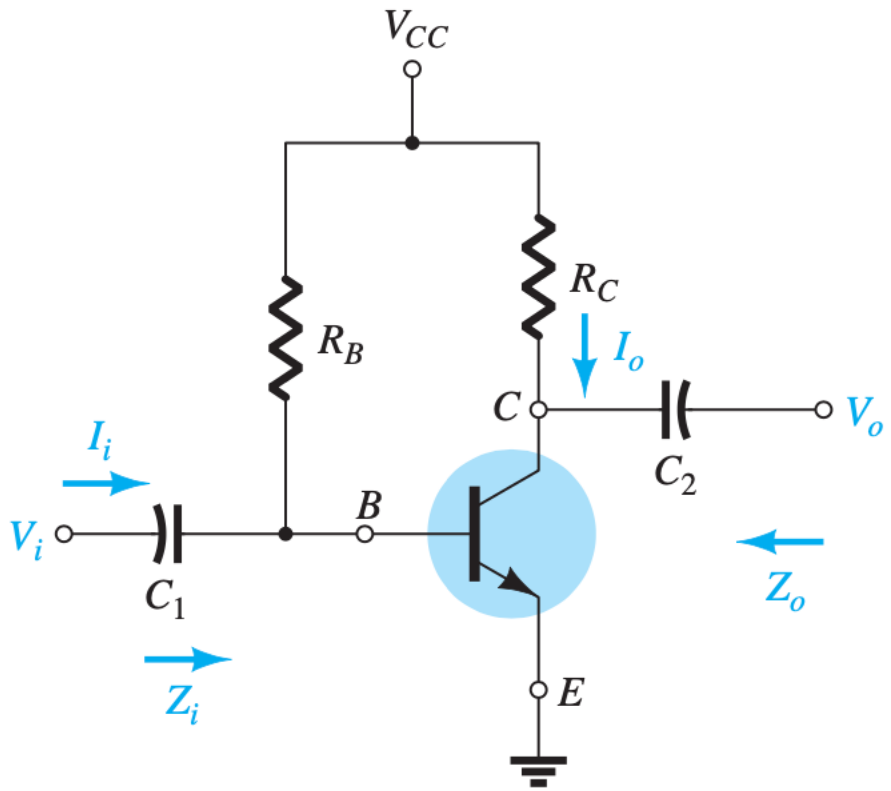
Electrical Characteristics (T_a = 25°C unless otherwise specified)

Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector Knee Voltage	$V_{CE\ (K)}$	I_C = 10mA, I_B = The value for which I_C = 11mA at V_{CE} = 1V	-	0.60	V
Transition Frequency	f_t	V_{CE} = 5V, I_C = 10mA, f = 100MHz	150	-	MHz
Noise Figure	NF	V_{CE} = 5V, I_C = 0.2mA R_g = 2kΩ F = 1kHz,B = 200Hz		10	dB
Output Capacitance	C_{obo}	V_{CB} = 10V, f = 1MHz	-	4.5	pF
Small Signal Current Gain	h_{fe}	ALL f = 1kHz I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	 125 125 125 240 450	 500 900 260 500 900	-
Input Impedance	h_{ie}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	 1.6 3.2 6.0	 4.5 8.5 15	 KΩ KΩ
Output Admittance	h_{oe}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	-	 30 60 110	umhos

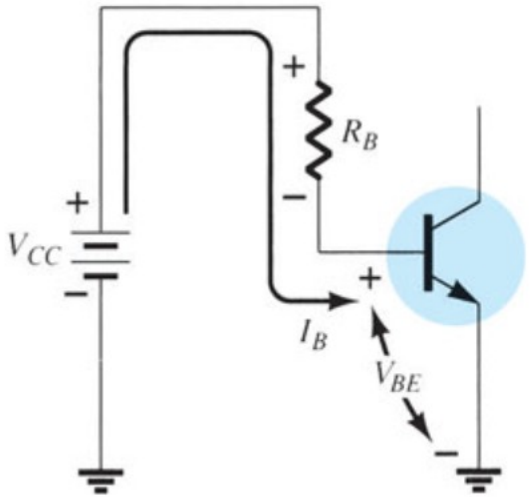
Kurva Karakteristik Output Common Emitter



Rangkaian Skematik Fixed Bias



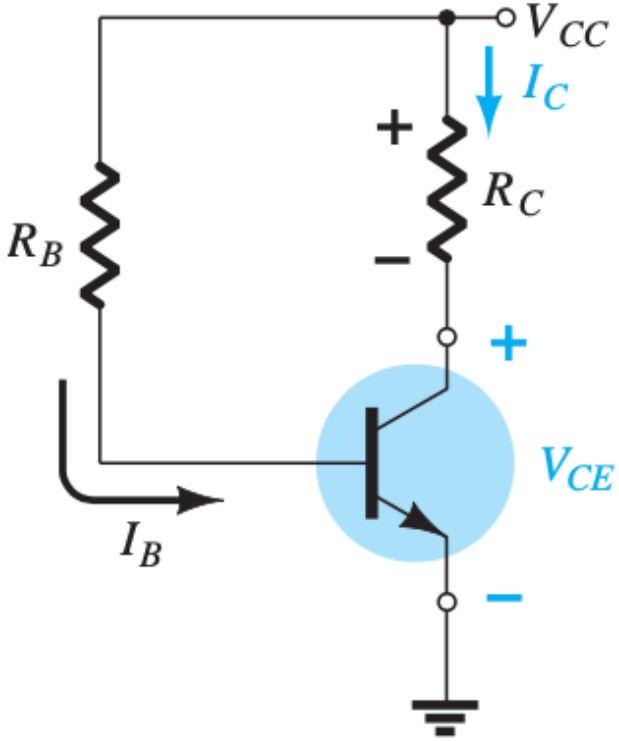
Analisis DC Fixed Bias

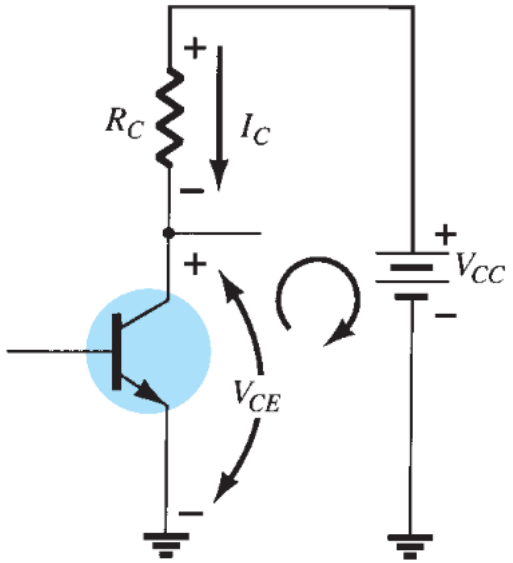


$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$





$$V_{CE} + I_C R_C - V_{CC} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

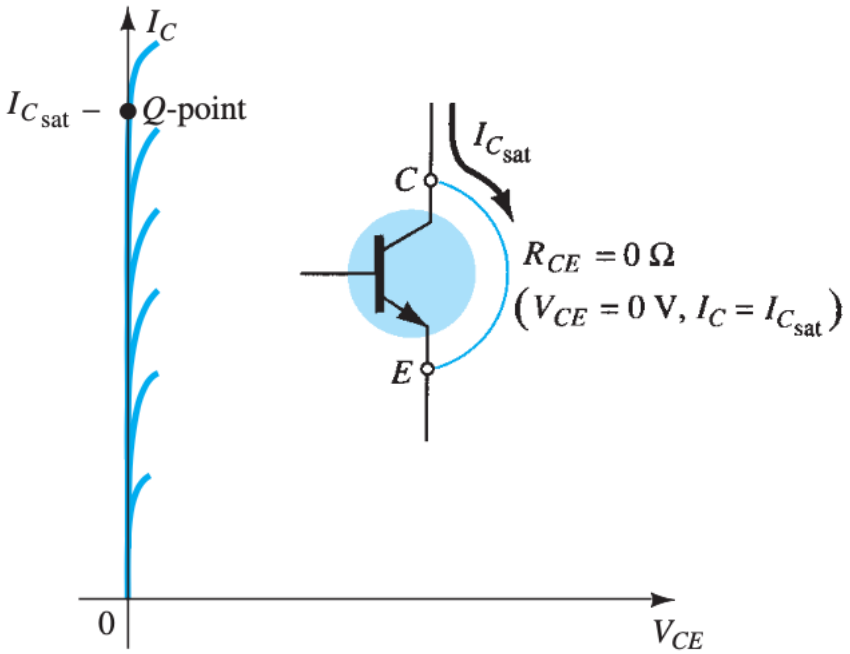
Menentukan Arus Collector Maksimum

Transistor Saturation

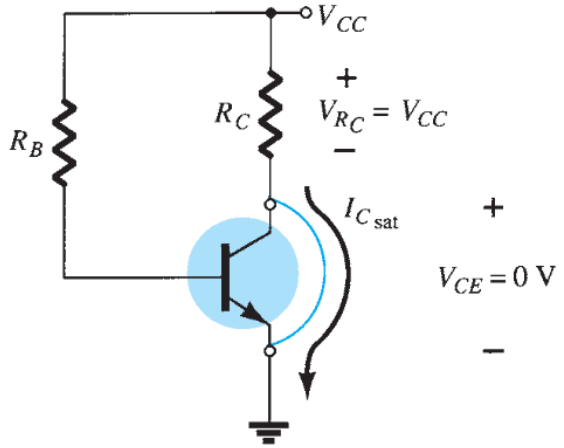
The term *saturation* is applied to any system where levels have reached their maximum values.

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\ \text{V}}{I_{C_{sat}}} = 0\ \Omega$$

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Approximate



PENTING!!!!

Setelah I_{Csat} diketahui, kita mendapatkan gambaran arus collector maksimum untuk desain yang digunakan. Jika ingin mendapatkan amplifikasi linier, maka titik kerja arus collector harus berada di bawah level I_{Csat}

Garis Beban dan Penentuan Titik Kerja

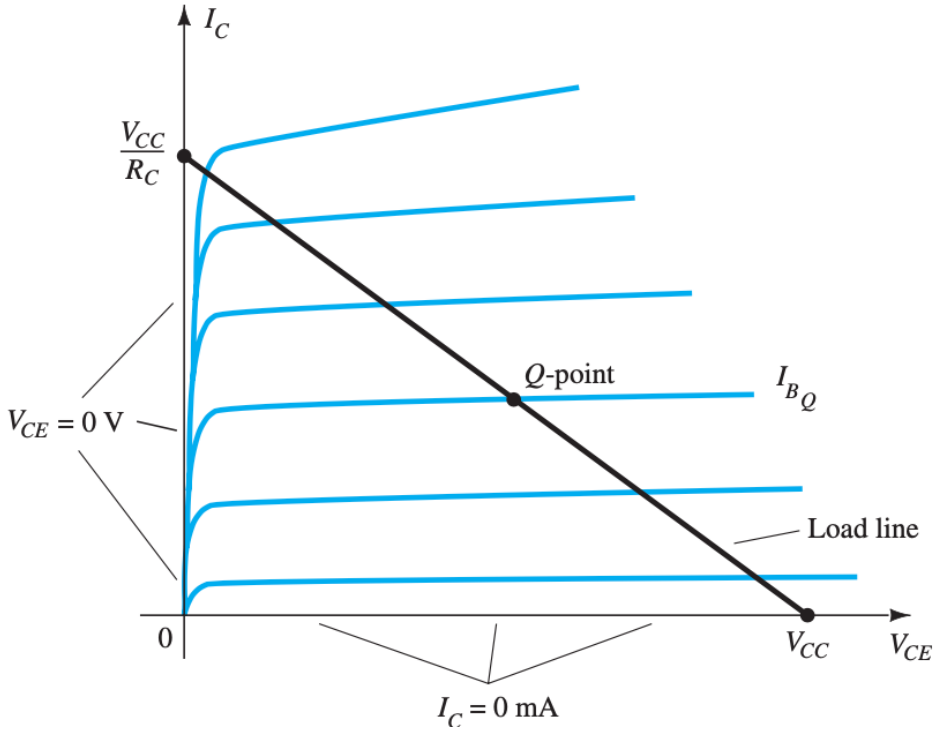
$$V_{CE} = V_{CC} - I_C R_C$$

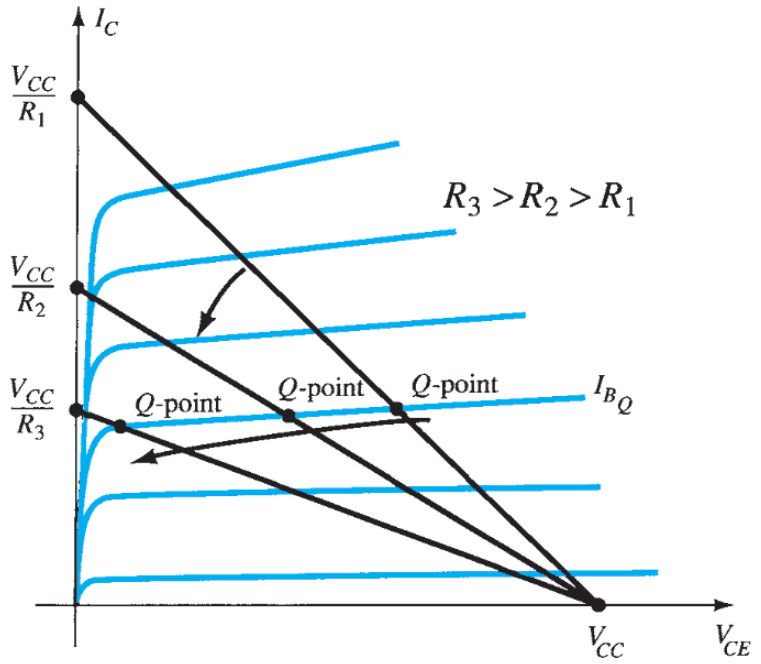
$$V_{CE} = V_{CC} - (0)R_C$$

$$V_{CE} = V_{CC}|_{I_C=0\text{ mA}}$$

$$0 = V_{CC} - I_C R_C$$

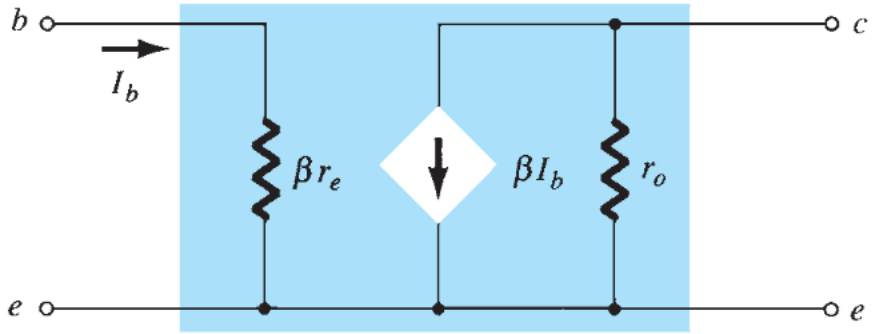
$$I_C = \frac{V_{CC}}{R_C}\bigg|_{V_{CE}=0\text{ V}}$$





Pengaruh nilai R terhadap Garis Beban

Rangkaian Ekivalen BJT Common Emitter



$$r_e = 26\,\text{mV}/I_E$$

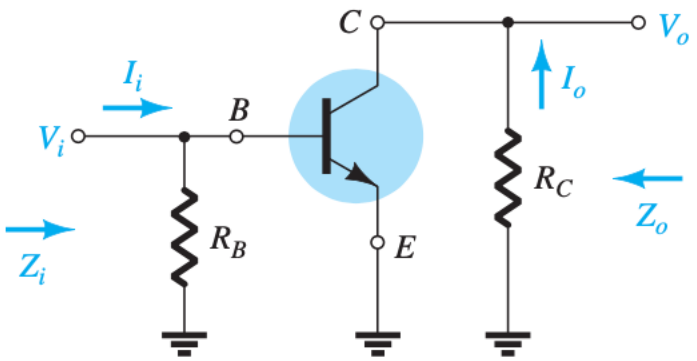
$$Z_i = \frac{V_i}{I_b} = \frac{V_{be}}{I_b}$$

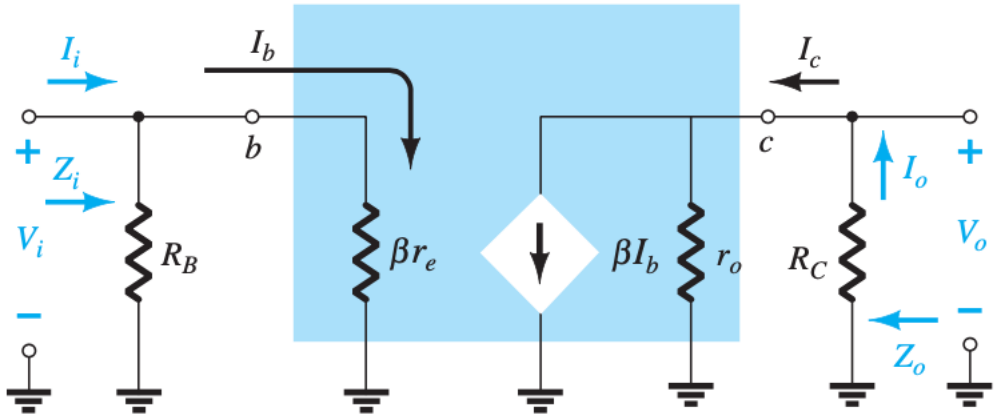
$$V_{be} = I_e r_e = (I_c + I_b) r_e = (\beta I_b + I_b) r_e$$
$$= (\beta + 1) I_b r_e$$

$$Z_i = \frac{V_{be}}{I_b} = \frac{(\beta + 1) I_b r_e}{I_b}$$

$$\boxed{Z_i = (\beta + 1) r_e \cong \beta r_e}$$

Analisis AC Fixed Bias





$$Z_i = R_B \| \beta r_e \quad \text{ohms}$$

$$Z_i \cong \beta r_e \quad R_B \geq 10\beta r_e \quad \text{ohms}$$

$$V_o = -\beta I_b (R_C \| r_o)$$

$$I_b = \frac{V_i}{\beta r_e}$$

$$V_o = -\beta \left(\frac{V_i}{\beta r_e} \right) (R_C \| r_o)$$

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \| r_o)}{r_e}$$

$$A_v = -\frac{R_C}{r_e} \quad r_o \geq 10R_C$$

Diagram wiring pengukuran (simulasi) untuk analisis DC

V_{CC} = 10 V
I_{Csat} = 10mA
R_C=1K
I_{CQ} = 5mA
β =292
I_B = 16.7uA
R_B = 556886 = 560K Ohm

